

Predictive risk factors for radiation retinopathy and optic neuropathy after proton beam therapy for uveal melanoma

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Abstract

Purpose This study was performed in order to evaluate the incidence of radiation retinopathy and optic neuropathy occurring after proton beam therapy for uveal melanoma.

Methods Included in this study were all patients who had been treated with primary proton beam therapy for uveal melanoma at the oncology service between May 1998 and June 2014 with a minimum follow-up of 12 months. Excluded were all patients who underwent re-irradiation, or vitrectomy due to exudative retinal detachment or for tumor-resection.

Results During this period, 1127 patients matched the inclusion criteria, of whom 768 (68.1 %) and 463 (41.0 %) developed radiation retinopathy and optic neuropathy after a median time of 18.9 months (2.0–99.84 months) and 19.8 months (0.2–170.4 months), respectively. Mean follow-up was 53.4 months (12–170.4 months). Included were 558 men (49.5 %) and 569 women (50.5 %). Mean age was 61 years (16–89 years). Visual acuity slightly decreased from initial levels of 0.3 logMAR–0.4 logMAR in patients without developing any radiation-induced complication but severely decreased to 1.0 logMAR or 1.5 logMAR in the case of developing radiation retinopathy only or optic neuropathy, respectively. Independent risk factors for radiation retinopathy were a centrally (<2.5 mm from sensitive structures) located tumor

or a thick tumor located more than 2.5 mm from sensitive structures, while those for radiation optic neuropathy comprised a short distance and applied dose to the optic disk.

Conclusion The risk for radiation retinopathy is higher in central uveal melanoma.

Mid-/peripheral tumors are at high risk for radiation retinopathy and maculopathy if presenting with increased thickness.

Keywords Radiation retinopathy · Proton beam therapy · Tumor resection · Uveal melanoma

Introduction

Radiation retinopathy or optic neuropathy are the main limiting factors for visual outcome after irradiation treatment of uveal melanoma. This study concentrates on the radiation-induced complications after proton beam therapy, while aiming to detect risk factors related to tumor characteristics and irradiation doses. The aim of this study is to therefore identify patients at high risk for complications requiring modified post-irradiation care.

Material and methods

The study was approved by the institutional review board of the Charité-Universitätsmedizin, Berlin, Germany, in accordance with the tenets of the Declaration of Helsinki. It is a retrospective, consecutive, single-institution case series that included 1127 patients with choroidal-/ciliary body-melanoma who had received a 60 Cobalt Gray Equivalent (CGE) proton beam therapy as a primary treatment at the Department of Ophthalmology of Charité

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Universitätsmedizin Berlin and the Helmholtz Zentrum Berlin between January 1998 and June 2014. In our oncology service, the main indication for proton beam therapy is a tumor located within the arcades or one that exceeds 6 mm in height. Melanoma up to 6 mm or peripheral tumors beyond the arcades are treated with ruthenium brachytherapy.

Exclusion criteria for this study were re-irradiation for local recurrences, surgical resection, and/or vitrectomy for exudative retinal detachment. Minimum follow-up was 12 months after proton beam therapy. The data was obtained from electronic health records, operation reports, discharge letters, and radiation planning protocols. The dose of proton beam therapy in our study was 60 CGE in total (Cobalt Gray Equivalent: 1 CGE = 1.1 Gray (Gy), taking a radio biological effectiveness of 1.1 into account), given in four fractions of 15 CGE at 4 sequential days [1]. Dose calculation and treatment planning were performed with the treatment planning systems EYEPLAN version 1.2 and since 2006 OCTOPUS (OCular TumOur Planning UtilitieS) version 4.4 [2, 3]. Distances between tumor and optic disc, fovea, and equator were measured on the retinal surface of the eye model generated by the treatment planning system. The eye model and the tumor are reconstructed from all available data: X-ray imaging of the eye with the tantalum clips, computer tomography, magnetic resonance imaging, ultrasound, optical coherence tomography, and fundus pictures.

Radiation retinopathy was diagnosed clinically as a new onset of retinal capillary changes including microaneurysms or dilation of retinal vessels, cotton wool spots, hemorrhages, and exudates [4]. Radiation optic neuropathy was defined as disc edema, radiation-induced disc pallor, parapapillary hemorrhages or exudates, or optic nerve fiber layer infarction [4]. Radiation maculopathy comprised patients presenting with macular edema associated with radiation retinopathy. Diagnosis was made clinically by ophthalmoscopy and, when available, confirmed by optical coherence tomography.

Statistical analysis

All examination results and findings were statistically evaluated using SPSS (Statistical Package for Social Sciences) for Windows, Release 22.0.

Comparisons between groups were performed by Mann–Whitney-U test, Kruskal-Wallis test, or independent-samples T-test. A statistical significance was set at $p < 0.05$. The factors detect ased significant on univariate analysis were considered for multivariate analysis using a forward stepwise model. Survival estimates were generated on the basis of the Kaplan-Meier method.

Results

Of 1127 patients, 1070 (94.9 %) patients presented with choroidal melanoma and 57 (5.1 %) patients with choroidal ciliary body melanoma. Patients were observed for a median of 46.2 months (mean 53.4 months) (range 12–170.4 months). There were 558 men (49.5 %) and 569 women (50.5 %). Mean age at baseline before proton beam therapy was 61 years (16–89 years).

General tumor characteristics: anatomical features and irradiation parameters

The median tumor thickness was 3.6 mm (0.1–14.1 mm), median largest basal diameter was 10.9 mm (3.0–22.4 mm), median distance to fovea was 0.6 mm (0–18.5 mm), median distance to optic disc was 1.6 mm (0–18.7 mm), and median distance to equator was 5.4 mm (0–18.0 mm). The median radiation dose to the fovea was 53 CGE (0–60 CGE) and that to the optic nerve was 6.0 CGE (0.60CGE). The optic nerve was irradiated for a median length of 0.1 mm (0–4.6 mm) and the ciliary body (divided in 12 sectors) was irradiated over 2.4 sectors (0–10.8 sectors).

Incidence of radiation-induced complications

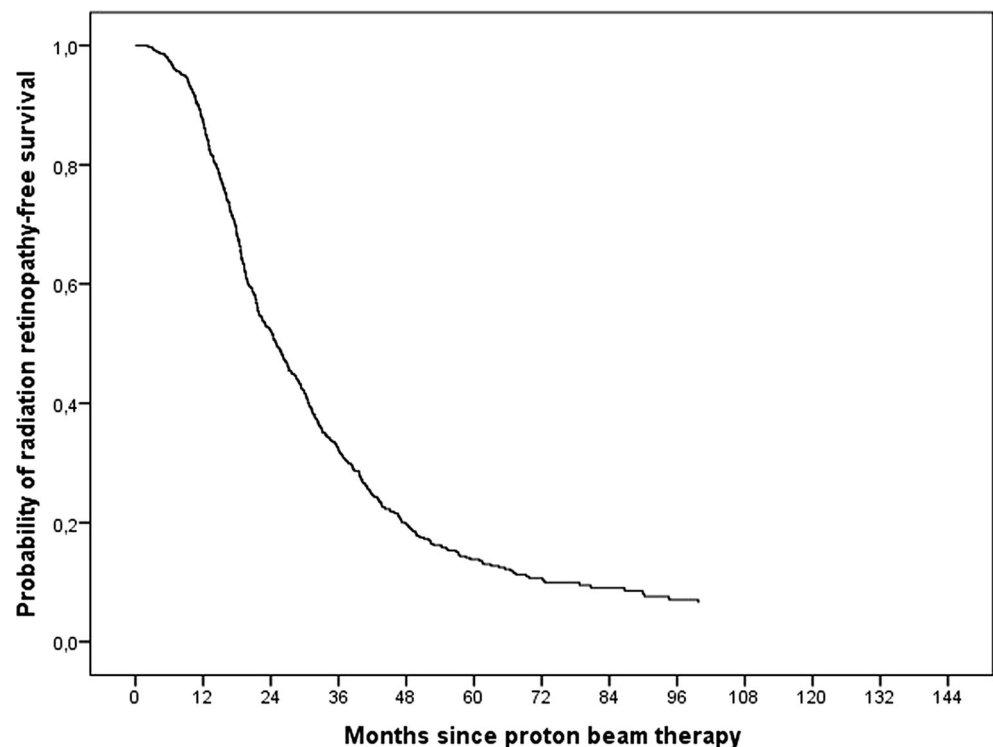
Of the 1,127 patients who met the inclusion criteria, 1,000 patients (88.7 %) developed combined radiation induced retinopathy and optic neuropathy. A total of 768 patients (68.1 %) developed only radiation retinopathy and 463 (41.0 %) developed isolated radiation optic neuropathy. Radiation retinopathy and optic neuropathy occurred after a median time of 18.9 months (2.0–99.8 months) and 19.8 months (0.2–170.4 months), respectively. The chances of being radiation retinopathy-free at 1, 2, 3, 4, 5, and 10 years after proton therapy were estimated to be 87.0, 53.0, 33.0, 21.0, 15.0, and 7 %, respectively. The chance of not having optic neuropathy following proton therapy was estimated to be 92 % after one year, 73.0 % after two years, 61.0 % after three years, 52.0 % after four years, 48.0 % after five years, and 26.0 % after 10 years. (Figs. 1, 2)

Information about radiation maculopathy was available in only 490 patients with radiation retinopathy. Of these 490 patients, 30 % (163 patients) developed radiation retinopathy-related macular edema.

Posterior segment complications: radiation retinopathy and optic neuropathy versus no complications, together with an analysis of predictive factors

In multivariate analysis, the occurrence of radiation retinopathy was independently influenced significantly ($p < 0.05$) by a centrally located tumor (large distance to equator, close

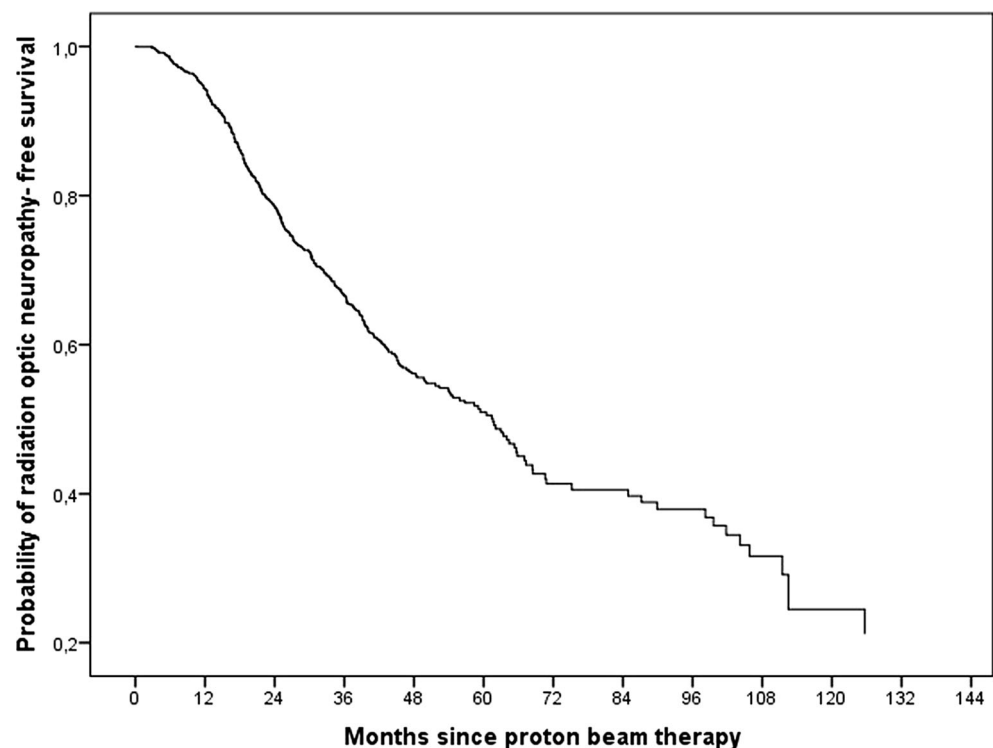
Fig. 1 All patients treated with proton beam therapy for uveal melanoma and their Kaplan-Meier analysis of survival without radiation retinopathy. The probabilities to survive radiation retinopathy-free for 1, 2, 3, 4, 5, and 10 years after proton therapy were estimated at 87.0, 53.0, 33.0, 21.0, 15.0, and 7 %, respectively



distance <2.5 mm to sensitive structures), tumor thickness, and a large basal diameter. Incidence of radiation optic neuropathy was significantly ($p < 0.05$) influenced by the proximity of the tumor to the optic disc and the dose applied to the optic disc. With increasing dose to optic disc, the incidence of

radiation-induced optic neuropathy increased: 16 % of patients receiving 0 CGE developed radiation optic neuropathy, 38.6 % developed radiation optic neuropathy with doses ranging from 1–20 CGE, 41 % with doses ranging from 21–40 CGE, 58.6 % with doses ranging from 41–50 CGE, and 72 %

Fig. 2 Kaplan-Meier analysis of survival without radiation optic neuropathy of all patients treated with proton beam therapy for uveal melanoma. The probability of an optic neuropathy-free survival was estimated at 92.0 % after 1 year, 73.0 % after 2 years, 61.0 % after 3 years, 52.0 % after 4 years, 48.0 % after 5 years, and 26.0 % after 10 years since proton therapy



receiving doses ranging from 51–60 CGE. As previously mentioned, from 490 patients, 163 patients (30 %) developed radiation retinopathy-related macular edema. Multivariate analysis identified a location close to fovea and a high dose on fovea as statistically significant risk factors leading to radiation-related edema, ($p = 0.02$ and $p = 0.05$).

Functional results

The median visual acuity prior to proton beam therapy was 0.3 logMAR, ranging from hand motions (HM) to 0.0 logMAR. At the last time of follow-up, the patients' median visual acuity was 1.3 logMAR, ranging from no light perception (NLP) to 0.0 logMAR.

Of 1127 patients, 127 patients did not develop any posterior segment complications; in this group, the initial median visual acuity was 0.3 logMAR and the final visual acuity was 0.4 logMAR (0.0–2.2 logMAR), while patients developing radiation-induced retinopathy and optic neuropathy had visual acuities of 0.3 logMAR before proton beam therapy and of 1.3 (0–2.2 logMAR) and 1.5 logMAR (0–2.2 logMAR) at most recent follow-up, respectively ($p = 0.001$). In total, 336 patients that presented with radiation retinopathy did not develop radiation optic neuropathy over the follow-up period. Their median initial visual acuity was 0.3 logMAR (HM to 0.0 logMAR) while final visual acuity was 1.0 logMAR.

Patients developing radiation retinopathy and macular edema had slightly worse final visual outcome, though not significant, with a median VA of 1.0 logMAR (mean 0.99 logMAR) compared to 0.9 logMAR (mean 0.94 logMAR) in patients without macular edema. There is inconsistent data due to patients having received treatment for macular edema (Table 1).

Subgroup analysis of midperipheral and peripheral tumors

For the subgroup analysis, only patients that presented with a tumor located further than 2.5 mm from the fovea and optic

disc and who did not receive radiation to those structures were included.

Of the 88 patients in this subgroup, 26 (30.6 %) developed radiation retinopathy after 14.9 months (12.0–55.6 months) and six patients also developed radiation optic neuropathy after 21.4 months (12.0–55.8 months).

The group of patients that developed radiation retinopathy differed significantly from those who did not in terms of tumor volume (820 mm³ versus 710 mm³), largest basal diameter (15.3 mm versus 13.1 mm), and tumor thickness (7.2 mm versus 6.1 mm) ($p < 0.01$).

Multivariate analysis revealed tumor thickness as an independent risk factor for radiation retinopathy (both $p < 0.001$).

The initial visual acuity of 0.3 logMAR in this subgroup was similar to the overall cohort but remained stable at 0.3 logMAR independent of whether the patient had developed radiation complications or not. Importantly, only five patients developed macular edema. These were all patients with tumors with a thickness greater than 11 mm and, in these five patients, the final visual acuity was reduced to 0.9 logMAR.

Discussion

In the present study, the total 5-year incidence rates for radiation retinopathy and optic neuropathy were calculated to be 85 and 52 %, respectively. These rates are higher than those reported in our subgroup analysis. Due to the study design, relatively smaller and more centrally located tumors were included, thereby complicating a direct comparison with other studies [5–10]. Furthermore, our results showed that a central location is an independent risk factor for radiation retinopathy. The same has previously been published by Gündüz et al., and has also been shown histopathologically [5, 11, 12]. In the present study, development of either of the two posterior segment complications was associated with a significant impairment of visual acuity, with optic neuropathy being the major contributor to loss of visual function, which has also been observed in previous studies [8, 9, 13, 14]. Subgroup analysis

Table 1 Visual acuity at initial and last examination according to radiation-induced complications for patients treated with proton beam therapy for uveal melanoma

	Initial visual acuity in logMAR <i>n</i> = 1,127	Final visual acuity in logMAR <i>n</i> = 1,127	Final visual acuity without radiation retino-/opticopathy in logMAR <i>n</i> = 127	Final visual acuity with radiation retinopathy (+/- -optic neuropathy) <i>n</i> = 768	Final visual acuity with radiation retinopathy solely in logMAR <i>n</i> = 336	Final visual acuity with radiation optic neuropathy in logMAR <i>n</i> = 463	Final visual acuity without radiation optic neuropathy <i>n</i> = 664
Mean	0.408	1.126	0.719	1.259	0.969	1.471	0.850
Median	0.300	1.3	0.4	1.3	1.0	1.5	0.700
Minimum	HM	NLP	NLP	NLP	NLP	NLP	NLP
Maximum	0.1	0.0	0.0	0.0	0.0	0.0	0.1

of midperipheral and peripheral tumors emphasized the significance of tumor thickness as the sole independent risk factor for radiation retinopathy. Thickness as a risk factor has previously been reported by different research groups as only being important in brachytherapy, which can be explained by requiring higher irradiation-doses in thicker tumors causing more damage to surrounding retina [5, 9, 15]. Additionally, equal results even after proton beam therapy with irradiation doses independent of tumor thickness (Bragg-peaks strung together receiving the Bragg-plateau) show that there must be some other explanation for this trend. This could most likely be owing to the fact that irradiated tumor tissue produces a high concentration of VEGF, leading initially to the so-called toxic tumor syndrome. Later in the clinical course, this can develop into radiation retinopathy and vision limiting macular edema [6, 16, 17]. Schönfeld et al. compared patients who underwent adjuvant endoresection with patients undergoing isolated proton beam therapy. In their study, it was found that reduced rates of neovascular glaucoma in the endoresection group occurred only after 6 years following proton beam therapy. This implies that the described neovascular glaucoma is not due to the toxic tumor syndrome but rather to radiation retinopathy [18]. Thus, tumor resections gain a new significance. Regarding the doses on the optic nerve head, no threshold dose could be determined, as was also observed histopathologically by Egbert et al. [11]. These results differ from those reported by Gragoudas et al., who reported a maximal tolerable dose of 30 CGE for optic neuropathy [19]. An explanation for this trend could be the individual variation in the blood supply of the optic disc, which consists of a variable network of posterior ciliary arteries, central retinal artery, and pial branches from other sources [20]. In conclusion, concerning the development of radiation maculopathy and radiation optic neuropathy, the strongest risk factor appears to be involvement and irradiation of the fovea and optic disc, respectively. The risk of developing radiation retinopathy is not only increased in more centrally located uveal melanoma (more or less regardless of dose), closer to sensitive structures, but also in thicker tumors. This evidence highlights the importance of tumor resections after radiation therapy in thicker tumors and close follow-up, including wide-field fluorescence angiography in central tumors.

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Compliance with ethical standards

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Conflict of interest statement Prof. A. Joussen and Dr. M. Rehak report personal fees from Bayer, personal fees from Novartis, and personal fees from Allergan, outside the submitted work. All other authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational

grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. For this type of study formal consent is not required.

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